

LI HAN

📍 Windsor, ON

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PROFESSIONAL SUMMARY

Professional with over eight years of experience in **embedded software verification, validation, and HIL systems** within the automotive industry, specializing in battery management systems (BMS), electric power steering (EPS), and other components. Demonstrates proven expertise in **system requirements analysis, test planning**, and a strong ability to **troubleshoot anomalies** and develop robust solutions. Skilled in **design analysis, cross-functional collaboration, and data-driven reporting**. A quick learner with excellent communication skills and a proactive approach to resolving technical issues.

TECHNICAL SKILLS

- **Systems & Domains:** BMS, EPS, Steering Wheel, Pre-pretensioner Mechatronic Integration (PPMI), Powertrain ECUs
- **Calibration & Validation:** CANape, INCA, CANoe, CANalyze, CANstress, VH6501, HIL, VT system
- **Test Engineering:** HIL Testing, Automation Testing, Fault Injection, Regression, Root Cause Analysis, DVP&R Planning
- **Software & Scripting:** VtestStudio, MATLAB/Simulink, CAPL, Python
- **Protocols & Standards:** CAN, LIN, UDS, ISO 26262, ASPICE
- **Collaboration Tools:** IBM DOORS, JIRA, SVN, PTC, PLM, IBM Rational Change, TestRail, Testlink

WORK EXPERIENCE

HIL Test Engineer

Sept. 2022 – Oct. 2024

SES Co., Ltd., Shanghai, China

- Reviewed and analyzed the software system requirements and developed test scope, plans, and test cases
- Analyzed the hardware design schematic and designed HIL hardware boards configuration
- Led commissioning and setup of HIL automation test systems
- Performed software functional and regression validation, generating reports
- Created anomalies in JIRA, supported root cause analysis, and resolved
- Developed automation scripts (Python, CAPL) to improve efficiency and reproducibility of testing

Achievements:

- ✧ Successfully commissioned the company's first HIL automation system
- ✧ Successfully completed the BMS software functional validation with high proficiency and quality

Senior Software Validation Engineer

Oct. 2021 – Sept. 2022

Autoliv Vehicle Safety Systems Technical Center Co., Ltd., Shanghai, China

- Reviewed customer requirements in DOORS, decomposed into functional test cases
- Commissioned VT test systems, created CAPL scripts and developed automation test cases
- Executed validation activities for steering wheel and PPMI
- Developed validation plans, created and tracked anomalies
- Coordinated cross-functional teams including PM, QA, and SDE

Achievements:

- ✧ Successfully completed the critical steering wheel software validation for the global project
- ✧ Mentored and trained team members on the new tools used for the automation system

Core Software Engineer

Aug. 2019 – Sept. 2021

Nexmeer Automotive System Co., Ltd., Suzhou, China

- Developed test cases based on software and customer specifications
- Executed EPS validation using VT system, CANoe and VtestStudio
- Delivered software release packages, including test results and compliance reports and provided data-driven recommendations to developers
- Collaborated with EPDT/software teams to troubleshoot anomalies and recommend robust solutions
- Led software validation activities, including designing test plans, coordinating test resources, and tracking test completion levels

Achievements:

- ✧ Successfully released over twenty software packages and supported at least five companies' important projects in mass production

Powertrain HIL Engineer

Mar. 2018 – Aug. 2019

Nanjing Future Mobility, Nanjing, China

- Set up and maintained BMS HIL test bench
- Conducted system requirements analysis and test cases design
- Assisted team members with vehicle integration validation on the test bench
- Collaborated with the global team members

Software Test Engineer

Apr. 2016 – Feb. 2018

CATL (Contemporary Amperex Technology Ltd.), Ningde, China

- Analyzed and managed system requirements in DOORS
- Designed test cases according to the requirements and features
- Performed full BMS function and regression tests, analyzed test data using CANoe, and managed test results with SVN
- Performed troubleshooting and reported software anomalies, supporting continuous improvement of system robustness
- Designed and reviewed test cases with team members
- Actively participated in customer workshops, presenting test insights and validation results

Achievements:

- ✧ Successfully completed software full functional validation for more than two important overseas customers' projects, as well as brought them to mass production

EDUCATION

Master of Engineering

2016 Completed

Harbin Engineering University, China

Bachelor of Engineering

2012 Completed

Heilongjiang University, China

TRAINING AND CERTIFICATIONS

Safe handling of high voltage systems in motor vehicles Certification | TUV

Apr., 2018

Electric Work in the Non-live State Certification | TUV

Oct., 2018

Live Electrical Work Certification | TUV

Jun., 2019